Section 3.1 IM-OLS Cointegration Estimation Matrix: U.S. NFC Transformation Elasticity (1929–2024)

Department of Economics, UMass Amherst

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Data Provenance and Estimation Summary

This report presents the empirical cointegration results for Section 3.1 of Chapter 2. All models are estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over the full NIPA sample window (1931–2024, $N = 94$ observations after listwise deletion for the lag-1 conditioning variables ω_{t-1} and ι_{t-1}). Point estimates are tuning-parameter free; coefficient and cointegration-test significance stars use Vogelsang & Wagner (2014) fixed-b critical values (Bartlett kernel, Newey-West bandwidth), interpolated by each model's bandwidth ratio and number of stochastic regressors – see the significance note under each table. Distributional conditioning variables (ω_{t-1} and ι_{t-1}) are centered at their 1929 initial values, preserving $\hat{\theta}_0$ as the capital elasticity at the initial state. VIF statistics are computed on the level regressors space. To ensure robustness against structural instability, each specification dynamically incorporates step level-shift dummies $D_{T_b}^{\text{step}}$ for structural breaks identified by a first-pass Bai-Perron residual audit.

1 Specification Set 1: Double Interactions ($k^{\text{cap}} \cdot \omega_{t-1}$)

Table 1: Specification Set 1: Double Interaction IM-OLS Cointegration Estimates (1929–2024)

Dependent Variable: y_t^{NFC} (Log NFC Real GVA)						
Model	M1.1	M1.2	M1.3	M1.4	M1.5	M1.6
Capital Stock	k_1^{cap}	k_2^{cap}	k_1^{cap}	k_2^{cap}	k_1^{cap}	k_2^{cap}
Demand Control	None	None	None	None	None	None
$\hat{\alpha}$ (Intercept)	8.96***	6.84***	9.19***	8.74***	5.42***	4.80***
	(0.83)	(0.78)	(0.91)	(0.82)	(1.14)	(1.15)
$\hat{\gamma}_{1944}^{\text{step}}$ (D_{1944}^{step})	0.57***	0.39**	—	0.58***	—	—
	(0.13)	(0.12)		(0.12)		
$\hat{\gamma}_{1945}^{\text{step}}$ (D_{1945}^{step})	—	—	0.56***	—	—	—
			(0.14)			
$\hat{\gamma}_{1965}^{\text{step}}$ (D_{1965}^{step})	0.39**	0.10	0.40**	0.36**	—	—
	(0.13)	(0.13)	(0.14)	(0.13)		
$\hat{\gamma}_{1967}^{\text{step}}$ (D_{1967}^{step})	—	—	—	—	—	-0.38*
						(0.15)
$\hat{\gamma}_{1968}^{\text{step}}$ (D_{1968}^{step})	—	—	—	—	-0.34	—
					(0.15)	
$\hat{\gamma}_{1981}^{\text{step}}$ (D_{1981}^{step})	—	—	—	—	—	-0.30
						(0.13)
$\hat{\gamma}_{1982}^{\text{step}}$ (D_{1982}^{step})	—	0.01	—	—	-0.27	—
		(0.13)			(0.13)	
$\hat{\gamma}_{1985}^{\text{step}}$ (D_{1985}^{step})	0.25	—	0.28	0.25	—	—
	(0.12)		(0.13)	(0.12)		
$\hat{\gamma}_{1996}^{\text{step}}$ (D_{1996}^{step})	—	—	—	—	0.12	0.07
					(0.10)	(0.10)
$\hat{\gamma}_{2003}^{\text{step}}$ (D_{2003}^{step})	0.19	—	0.18	0.16	—	—
	(0.16)		(0.16)	(0.15)		
$\hat{\gamma}_{2010}^{\text{step}}$ (D_{2010}^{step})	—	—	—	—	-0.31**	-0.37**
					(0.10)	(0.10)
$\hat{\theta}_0$ (Capital Elasticity)	0.34***	0.52***	0.33***	0.35***	0.67***	0.70***
	(0.07)	(0.07)	(0.08)	(0.07)	(0.10)	(0.10)
$\hat{\theta}_1$ (Interaction)	-0.05	-0.07	-0.07	-0.04	-0.20	-0.18
	(0.14)	(0.10)	(0.15)	(0.13)	(0.12)	(0.12)
Collinearity Diagnostics (VIF)						
VIF (k^{cap})	1.38	1.40	1.54	1.56	8.04	8.14
VIF (Interaction)	1.38	1.40	1.54	1.56	8.04	8.14
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.67***	-4.35***	-4.90***	-4.50***	-4.47***	-4.60***
Shin LM (C_{Shin}) [H_0: Coint.]	0.63***	1.02***	0.52***	0.71***	0.24***	0.22***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	5.53	802.81***	5.92	2750.38***	2585.04***	1472.40***
R^2_{sum}	1.00	1.00	1.00	1.00	1.00	1.00
Sample Size (N)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Coefficient rows ($\hat{\alpha}$, $\hat{\gamma}^{\text{step}}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual ADF and VW Type 2 rows use Vogelsang & Wagner (2014) fixed-b critical values, interpolated by each model's bandwidth ratio $b = \text{bandwidth}/T$ (Bartlett kernel, Newey-West bandwidth) and its number of stochastic regressors k – NOT a single generic t/normal cutoff. Resid. ADF stars instead follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8), which is required because the IM-OLS residuals are strongly autocorrelated – an unaugmented Dickey-Fuller is biased toward non-rejection and was the source of the superseded v2 results. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184); these have not been independently re-verified as regressor-count-specific for this report's $k = 2/k = 3$ mix and should be treated as provisional (see the audit memo referenced in the repository). **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0). Step dummies $D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$ represent persistent level shifts from structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming), distinguished from impulse outlier controls $D_{T_b}^{\text{imp}} = \mathbb{I}(t = T_b)$.

2 Specification Set 2: Triple Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1} \cdot \iota_{t-1}$)

Table 2: Specification Set 2 (Part A: Baseline NFC & Corporate GVA Wage Shares): Triple Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.1	M2.2	M2.3	M2.4	M2.5	M2.6
Capital Stock	k_1^{cap}	k_1^{cap}	k_2^{cap}	k_2^{cap}	k_1^{cap}	k_1^{cap}
Demand Control	None	NetC Share	None	NetC Share	None	NetC Share
$\hat{\alpha}$ (Intercept)	6.19***	7.39***	6.38***	7.89***	6.31***	7.33***
	(1.04)	(1.05)	(0.93)	(0.98)	(1.08)	(1.08)
$\hat{\gamma}_{1944}^{\text{step}}$ (D_{1944}^{step})	0.27	0.38	0.33*	0.55*	0.29	0.37
	(0.14)	(0.17)	(0.13)	(0.16)	(0.14)	(0.17)
$\hat{\gamma}_{1965}^{\text{step}}$ (D_{1965}^{step})	—	—	0.04	0.20	—	—
			(0.14)	(0.12)		
$\hat{\gamma}_{1966}^{\text{step}}$ (D_{1966}^{step})	0.00	0.09	—	—	0.02	0.09
	(0.16)	(0.13)			(0.16)	(0.13)
$\hat{\gamma}_{1981}^{\text{step}}$ (D_{1981}^{step})	—	-0.07	—	0.06	—	-0.07
		(0.14)		(0.12)		(0.14)
$\hat{\gamma}_{1982}^{\text{step}}$ (D_{1982}^{step})	-0.12	—	-0.06	—	-0.11	—
	(0.16)		(0.15)		(0.17)	
$\hat{\gamma}_{1998}^{\text{step}}$ (D_{1998}^{step})	—	—	—	0.27	—	—
				(0.10)		
$\hat{\gamma}_{1999}^{\text{step}}$ (D_{1999}^{step})	—	0.25	—	—	—	0.24
		(0.10)				(0.10)
$\hat{\theta}_0$ (Capital Elasticity)	0.59***	0.52***	0.56***	0.44***	0.58***	0.52***
	(0.09)	(0.08)	(0.08)	(0.07)	(0.09)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.09	0.88	0.01	0.52	-0.08	0.77
	(1.00)	(0.84)	(0.92)	(0.75)	(0.97)	(0.83)
$\hat{\beta}_{\text{cm}}$ ($\text{NetC}^{\text{share}}$)	—	-0.73	—	-0.33	—	-0.72
		(0.74)		(0.72)		(0.74)
Collinearity Diagnostics (VIF)						
VIF (k^{cap})	1.57	2.54	1.59	2.61	1.74	2.79
VIF (Interaction)	1.57	1.63	1.59	1.67	1.74	1.82
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.46***	-4.26**	-4.15**	-4.00*	-4.03**	-4.34**
Shin LM (C_{Shin}) [H_0: Coint.]	0.78***	0.77***	0.87***	0.67***	0.77***	0.77***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	1542.22***	19074.59***	1947.14***	21688.45***	2222.23***	19123.87***
R^2_{sum}	1.00	1.00	1.00	1.00	1.00	1.00
Sample Size (N)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Coefficient rows ($\hat{\alpha}$, $\hat{\gamma}^{\text{step}}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual ADF and VW Type 2 rows use Vogelsang & Wagner (2014) fixed-b critical values, interpolated by each model's bandwidth ratio $b = \text{bandwidth}/T$ (Bartlett kernel, Newey-West bandwidth) and its number of stochastic regressors k – NOT a single generic t/normal cutoff. Resid. ADF stars instead follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8), which is required because the IM-OLS residuals are strongly autocorrelated – an unaugmented Dickey-Fuller is biased toward non-rejection and was the source of the superseded v2 results. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184); these have not been independently re-verified as regressor-count-specific for this report's $k = 2/k = 3$ mix and should be treated as provisional (see the audit memo referenced in the repository). **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0). Step dummies $D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$ represent persistent level shifts from structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming), distinguished from impulse outlier controls $D_{T_b}^{\text{imp}} = \mathbb{I}(t = T_b)$.

Table 3: Specification Set 2 (Part B: Shaikh-Tonak Tier 2 Wage Share): Triple Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)						
Model	M2.7	M2.8	M2.9	M2.10	M2.11	M2.12
Capital Stock	k_2^{cap}	k_2^{cap}	k_1^{cap}	k_1^{cap}	k_2^{cap}	k_2^{cap}
Demand Control	None	NetC Share	None	NetC Share	None	NetC Share
$\hat{\alpha}$ (Intercept)	6.52***	7.03***	9.47***	10.35***	9.39***	10.04***
	(0.98)	(1.11)	(1.27)	(0.76)	(1.23)	(0.85)
$\hat{\gamma}_{1944}^{\text{step}}$ (D_{1944}^{step})	0.35*	0.43	0.54**	0.38*	0.54***	0.40**
	(0.13)	(0.16)	(0.14)	(0.11)	(0.13)	(0.11)
$\hat{\gamma}_{1965}^{\text{step}}$ (D_{1965}^{step})	0.06	—	—	—	—	—
	(0.15)					
$\hat{\gamma}_{1966}^{\text{step}}$ (D_{1966}^{step})	—	0.08	—	—	0.34	—
		(0.13)			(0.15)	
$\hat{\gamma}_{1967}^{\text{step}}$ (D_{1967}^{step})	—	—	0.34	0.02	—	0.01
			(0.16)	(0.08)		(0.09)
$\hat{\gamma}_{1981}^{\text{step}}$ (D_{1981}^{step})	—	-0.05	—	0.04	—	0.05
		(0.13)		(0.09)		(0.10)
$\hat{\gamma}_{1982}^{\text{step}}$ (D_{1982}^{step})	-0.04	—	0.10	—	0.14	—
	(0.15)		(0.15)		(0.14)	
$\hat{\gamma}_{1996}^{\text{step}}$ (D_{1996}^{step})	—	—	—	-0.16	—	-0.16
				(0.08)		(0.08)
$\hat{\gamma}_{1998}^{\text{step}}$ (D_{1998}^{step})	—	0.23	—	—	—	—
		(0.10)				
$\hat{\gamma}_{2010}^{\text{step}}$ (D_{2010}^{step})	—	—	—	-0.49***	—	-0.50***
				(0.09)		(0.09)
$\hat{\theta}_0$ (Capital Elasticity)	0.55***	0.52***	0.30*	0.31***	0.30*	0.32***
	(0.08)	(0.08)	(0.11)	(0.06)	(0.11)	(0.07)
$\hat{\theta}_1$ (Interaction)	0.00	0.87	-2.25*	-5.58***	-2.01	-5.34***
	(0.90)	(0.78)	(0.90)	(0.94)	(0.88)	(0.97)
$\hat{\beta}_{\text{cm}}$ ($\text{NetC}^{\text{share}}$)	—	-0.52	—	-1.66**	—	-1.50*
		(0.71)		(0.44)		(0.45)
Collinearity Diagnostics (VIF)						
VIF (k^{cap})	1.78	2.88	6.76	9.56	7.09	10.32
VIF (Interaction)	1.78	1.87	6.76	7.53	7.09	8.07
Cointegration Test Triad						
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-3.79**	-4.09*	-3.89**	-5.44***	-3.95**	-5.66***
Shin LM (C_{Shin}) [H_0: Coint.]	0.85***	0.70***	0.52***	0.12**	0.50***	0.13**
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	2740.33***	19999.66***	202.86***	36650.60***	161.62***	36740.01***
R^2_{sum}	1.00	1.00	1.00	1.00	1.00	1.00
Sample Size (N)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Coefficient rows ($\hat{\alpha}$, $\hat{\gamma}^{\text{step}}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual ADF and VW Type 2 rows use Vogelsang & Wagner (2014) fixed-b critical values, interpolated by each model's bandwidth ratio $b = \text{bandwidth}/T$ (Bartlett kernel, Newey-West bandwidth) and its number of stochastic regressors k – NOT a single generic t/normal cutoff. Resid. ADF stars instead follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8), which is required because the IM-OLS residuals are strongly autocorrelated – an unaugmented Dickey-Fuller is biased toward non-rejection and was the source of the superseded v2 results. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184); these have not been independently re-verified as regressor-count-specific for this report's $k = 2/k = 3$ mix and should be treated as provisional (see the audit memo referenced in the repository). **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0). Step dummies $D_{T_b}^{\text{step}} = \mathbb{I}(t \geq T_b)$ represent persistent level shifts from structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming), distinguished from impulse outlier controls $D_{T_b}^{\text{imp}} = \mathbb{I}(t = T_b)$.